

Metabolic compartmentation

Mitochondria	Fatty acid oxidation (β -oxidation), acetyl-CoA production, TCA cycle, oxidative phosphorylation, ketogenesis.
Cytoplasm	Glycolysis, HMP shunt, and synthesis of cholesterol (SER), proteins (ribosomes, RER), fatty acids, and nucleotides.
Both	Heme synthesis, urea cycle, gluconeogenesis. Hugs take two (both).

Summary of pathways

- 1 Galactokinase (*mild galactosemia*)
- 2 Galactose-1-phosphate uridylyltransferase (*severe galactosemia*)
- 3 Hexokinase/glucokinase
- 4 Glucose-6-phosphatase (*von Gierke disease*)
- 5 Glucose-6-phosphate dehydrogenase
- 6 Transketolase
- 7 Phosphofructokinase-1
- 8 Fructose-1,6-bisphosphatase 1
- 9 Fructokinase (*essential fructosuria*)
- 10 Aldolase B (*fructose intolerance*)
- 11 Aldolase B (*liver*), A (*muscle*)
- 12 Triose phosphate isomerase
- 13 Pyruvate kinase
- 14 Pyruvate dehydrogenase
- 15 Pyruvate carboxylase
- 16 PEP carboxykinase
- 17 Citrate synthase
- 18 Isocitrate dehydrogenase
- 19 α -ketoglutarate dehydrogenase
- 20 Carbamoyl phosphate synthetase I
- 21 Ornithine transcarbamylase
- 22 Propionyl-CoA carboxylase
- 23 HMG-CoA reductase

